Madhav Raj

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Summary

Computer Science undergraduate at NIT Delhi with hands-on experience in full-stack development, generative AI, and DSA. I have a track record of building production-grade applications and AI-driven systems including a multi-accent voice agent at Accenture and a campus ridesharing platform serving 200+ users.

Education

National Institute of Technology Delhi **Aug 2023 - May 2027**
B.Tech in Computer Science and Engineering (CSE) *Delhi, India*

Experience

Accenture – Advance Application Engineering Intern **June 2026 – July 2026**
Generative AI, Python, React Native *On-site*

- Developed **ARIA**, a multi-accent voice agent with accuracy across **5 distinct global English accents**, as part of Accenture's generative AI and agentic AI initiatives.
- Engineered a machine learning pipeline integrating **Faster-Whisper (STT)**, **Llama 3 (LLM)**, and **F5-TTS**; utilized bidirectional **WebSockets** for audio streaming, achieving end-to-end responses.
- Implemented a fine-tuned **Wav2Vec2 model** and successfully processing and classifying over **100+** audio interactions with **90% accent detection confidence**.
- Designed a high-performance, **60-FPS** frontend interface featuring custom HTML5 Canvas particle physics for state visualization, optimizing React state management.

Skills

Programming Languages & Databases: C++, Python, JavaScript, SQL

Libraries & Tools: React, React Native, Node.js, Pandas, NumPy, Scikit-learn, Git/GitHub, Supabase

Soft Skills: Communication, Project Management, Leadership, Teamwork

Projects

UniPool – Campus Ridesharing App **Live**
React Native, Expo, Supabase, TypeScript, Zustand *February 2026 – May 2026*

- Developed a cross-platform mobile application using React Native (Expo) and Typescript to connect people heading to the same destination and serving **200+ campus users**.
- Engineered a frictionless ride-creation flow enabling drivers to instantly post routes, set seat availability, split fares, and share rides across platforms, reducing average booking time by **40%**.
- Optimized state management using **Zustand** and **TanStack Query** to cache server requests, cutting network overhead by **30%**, and integrated AsyncStorage for offline preference persistence.

Bot Detection System – Walmart Sparkathon **GitHub**
Python, FastAPI, Machine Learning

- Designed a real-time ML system to detect and block malicious bot purchases during high-demand holiday sales, achieving **94% detection accuracy** in simulations.
- Simulated attacks with a self-built bot and reduced false positive rates by **40%**.

MediCare Fraud Analysis Software **GitHub**
Python, XGBoost, Machine Learning, Web Development

- Deployed an ML system to detect fraudulent Medicare providers using public datasets, achieving **91% accuracy**.
- Applied data cleaning, feature engineering, and trained an XGBoost model with high precision on fraud detection.

Achievements & Certifications

Credly Profile – Credly

Google Cybersecurity Professional Certificate V2 – *Certificate*

Finalist – Vishleshan'24 Annual Flagship Case Competition (VNIT Nagpur) – *Certificate*

Solved 300+ DSA and SQL questions and 1800+ rating on Leetcode

Extracurricular and Coursework

Extra-curricular Activities: Participated in various case study competitions and hackathons via Unstop and Hack2Skill platforms; **Executive member** of GDSC club at NIT Delhi; player in college football team.

Courses: Object-Oriented Programming, Data Structures & Algorithms, Database Management Systems, Operating Systems, Artificial Intelligence, Software Engineering, Computer Vision, Information Security, App Development